

# Choupo

The Open Source Chemical Process Simulator

# Explorer Guide

Choupo-dev

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# 1 What the Explorer is

The Explorer is Choupo's *interactive thermodynamics bench*: pick components, pick a view, and SEE what the selected property models do — instantly, in the browser, with no case authoring at all. It answers the question that comes *before* any simulation: **"what does this thermodynamics look like?"**

It sits deliberately between two other surfaces:

- The **props bench** (`choupoProps` cases) asks the same questions *reproducibly*: a case file, golden-locked numbers, validation blocks. The Explorer is its sketchpad — what you find here, you pin down there.
- The **simulators** consume the models the Explorer displays. A surprise in a flowsheet (an azeotrope, a scaling mineral, a runaway $\gamma$) is usually visible in the Explorer in thirty seconds.

Everything the Explorer draws is computed by the SAME C++ engine (compiled to WebAssembly) with the SAME curated catalogue (`data/standards/`) the solvers use. There is no "display model": if the Explorer shows an azeotrope at $x = 0.89$, the distillation column will hit it at $x = 0.89$.

# 2 The workflow

Explore is *one* tab with two **views** of one selected set, switched in place by the **Catalogue | Surfaces** control at the top left: the *Catalogue* browses the substances and reads their records, the *Surfaces* plot them. The set and the view travel in the address (`?workspace=explore&view=surfaces&components=water,ethanol`), so an exploration is a link you can hand to somebody.

1. **Pick components** in the left panel. The list is the live catalogue; each card shows the data the component actually carries (criticals, $P^{\mathrm{sat}}$, $c_p$, formation — and what it lacks). Case-local components (from an open case's `constant/`) appear alongside, marked.
2. **Pick a view** (the tabs enable themselves when their component-count and data requirements are met; a disabled tab tells you WHY it is disabled).
3. **Pick the method** where more than one applies — activity model, equation of state, $P^{\mathrm{sat}}$ correlation. Overlaying two methods on the same axes is the fastest modelling lesson Choupo offers: the gap between the curves IS the modelling uncertainty.
4. **Export**: every view is a Plotly figure (zoom, hover, save PNG) backed by numbers you can carry into a props case for the golden-locked version.

# 3 The views, one by one

## 3.1 The Catalogue view (the compound browser)

The catalogue is a *tree*, not a list. Groups derived from each record's own facts fold and unfold with their member count; the groups a student charges to a vessel open by default, the combustion library, the radicals and the synthetic test stand-ins start folded. A group too large for one screen is subdivided into *families*, and the family comes from the record, never from the name: first the *declared UNIFAC groups* (the highest-precedence group present decides — an acid before an ester before an alcohol, down to an alkane); only for a record with no group decomposition, the *elemental formula*, coarser and labelled so with a "(by formula)" suffix. A record with neither is shown under *"No formula declared in the record"* — a visible curation gap, never a guess. Typing in the search box flattens the tree; a double click inspects a component without changing the set. The browser reads the catalogue; it does not edit a record.

The browser is the left half of the *Catalogue* view — the view `?workspace=explore` opens when the address names no set. A click adds a compound to the set *and* shows its record in the right half — identity, what the record carries, the curation dossier's verdict — so the panel follows your reading; a double click reads a record without changing the set. **Plot $n$ compounds** switches to the *Surfaces* view in place, and for a pair some study has measured, **Evidence for this pair** opens the Evidence view (§3.15). The Surfaces do not need the Catalogue: they keep a compound rail of their own and open perfectly well with nothing selected (`?workspace=explore&view=surfaces`). A component the address names but the catalogue cannot resolve is left out of the set and said so, never dropped in silence. **F1** on the Catalogue view opens this section; on the Surfaces it opens the Theory Guide section of the active lens. The old addresses — `?workspace=properties` and `?workspace=compounds` — still work: they are rewritten to the view they used to open.

## 3.2 Property vs $T/P$ (scan)

Any pure-component property — $P^{\mathrm{sat}}$, liquid density, $c_p$, viscosity, enthalpy — swept over temperature or pressure, for one or many components. Use it to: check a correlation's range before trusting it in a case; compare components; spot the nonsense that extrapolation produces outside a fit's window (the guide's recurring lesson: a correlation is DATA, and data has a domain).

## 3.3 Pure phase diagram ($P$–$T$)

The full phase map of one substance: sublimation, melting and vapour-pressure lines meeting at the triple point, ending at the critical point. Expect water's negatively-sloped melting line (the anomaly), and the supercritical region where the liquid/vapour distinction dissolves.

## 3.4 Binary boiling envelope ($T$–$x$–$y$)

The bubble and dew curves of a binary at fixed $P$. This is where azeotropes announce themselves: the curves touch, and no distillation can cross the touching point. Switch the activity model (ideal → NRTL → UNIFAC) and watch the envelope bend — ideal solutions cannot azeotrope; real ones do.

## 3.5 $\gamma(x)$ — activity coefficients

The raw non-ideality of a binary: $\gamma_1, \gamma_2$ against composition, with the infinite-dilution values $\gamma^{\infty}$ at the edges. $\gamma > 1$: the pair dislikes mixing (positive deviation, minimum-boiling azeotropes); $\gamma < 1$: association (maximum-boiling). The Gibbs–Duhem constraint ties the two curves together — one cannot bend without the other answering.

## 3.6 Binary flash ($x$–$y$ + lever rule)

One flash, drawn: feed, vapour and liquid on the equilibrium diagram with the lever rule made visible. Use it to understand what a single equilibrium stage can and cannot do.

## 3.7 Binary and ternary LLE

Liquid–liquid splitting from the mixing Gibbs energy: $g_{\mathrm{mix}}(x)$ with the common-tangent construction (binary), and the ternary solubility envelope with tie-lines (UNIFAC-predicted). Expect the tangent points — not the minima — to define the coexisting phases; that distinction is the whole thermodynamics of demixing.

## 3.8 Ternary boiling surface

$T_{\mathrm{bubble}}$ over the composition triangle: valleys are distillation boundaries, and azeotropes sit at the stationary points.

## 3.9 Scaling (SI vs recovery)

The membrane/brine screening view: concentrate a water composition over recovery and watch each mineral's saturation index climb toward SI = 0 (the precipitation line). The Davies-vs-Pitzer method switch is the lesson: at high ionic strength the two activity models give DIFFERENT scaling verdicts, and choosing one is an engineering decision.

## 3.10 Cohesion of the pure liquids (Hildebrand parameter)

One bar per *pure* liquid at the stated temperature: $\delta = \sqrt{(\Delta H_{\mathrm{vap}}(T) - RT)/V_m}$, derived from each record's own $\Delta H_{\mathrm{vap}}$, $T_c$ and $V_{\mathrm{liq}}$ and never stored. A larger value is a more cohesive liquid, *not* a larger dissolving power, and no mixture property is computed in this view. The cohesive energy includes hydrogen bonding; what the single parameter cannot do is separate it from dispersion and polarity, so two close values make a pair a *candidate*, never a verdict — water and ethanol sit 22 $\mathrm{MPa}^{1/2}$ apart and mix in all proportions. The three-parameter Hansen split is what that needs: the engine reads it from a `hansen { dD; dP; dH; }` block on the component record and draws the Teas coordinates, but no shipped record carries one yet, so what is missing is licence-clean *data*, not code — the view and the run say so rather than inventing a triple. The `published` column is an independently reported value carried as an anchor that checks the derivation and never feeds it. **F1** opens the Theory Guide's chapter on the solubility parameter (`ch:solubility-parameter`), which derives the number and the Teas triangle and states both limits.

## 3.11 Equilibrium map (Gibbs)

The chemical-reaction view: pick two or more gas-phase species (with plain formulas — the atom matrix is derived from them, glass-box) and a feed, and get iso-lines of a product's equilibrium mole fraction over the $T \times P$ plane. The whole point is the tension it makes visible: for ammonia ($N_2 + 3H_2 \rightleftharpoons 2NH_3$) the equilibrium optimum sits cold and high-pressure, yet the labelled industrial window sits hot — where the catalyst can work. Answers the beginner's real confusion first ("where do I write the reaction?" — you don't; Gibbs redistributes the ATOMS you fed, shown as a live element balance). *Click any cell* for its full equilibrium composition and the ready-to-run `gibbsReactor` case dict — the map is a case generator. The *advanced* toggle exposes the temperature-approach $\Delta T$ (the empirical closeness-to-equilibrium knob commercial tools hide); it snaps rather than animates, and ghosts the $\Delta T = 0$ contours underneath so you read the before/after, not a motion. Unconverged cells are marked, never interpolated.

## 3.12 Steam tables (IF97)

Saturation and superheat properties of water from IAPWS-IF97 — $h_f, h_g, h_{fg}$, entropies, volumes — the backbone of every steam and power-cycle case, browsable.

## 3.13 EduTools (the Methods workspace)

Since 2026-08-15 the classical *method constructions* live in their own workspace, **EduTools**, split from the Explorer by one criterion: *a property surface answers "what does this system's property look like?"* (the Explorer); *a method construction answers "what does the classical graphical method say about a design question?"* — McCabe–Thiele, Kremser, Rayleigh, $\varepsilon$-NTU, pinch composites, and derivation pages for the property models. Open it from the hub's **EduTools** button: it is a tab of its own, needs no case, and does not talk to the flowsheet. A tool is chosen from the top row of the tab; every tool carries a one-line "teaches" caption and its Theory Guide link. Every curve is an engine run (`choupoProps` in WebAssembly) over a transient synthesized case — *zero physics in TypeScript*; the staircase drawn on it is geometry. Planned tools stay visible and disabled, naming what will feed them. It never writes a case: "Author → copy propsDict" emits the dict for you to author on disk.

## 3.14 Literature ("who measured this?")

Provenance is the differentiator, and this workspace puts its question one click from the flowsheet. With a case open, the tab's menu carries **Literature**: type compound names (every word must match a compound in the article) or click a chip for one of the case's own components, and read who measured that system — authors, year, title, journal, a DOI link, and the properties measured. It searches your *private* mirror of the NIST/TRC ThermoML Archive, installed once with `bin/choupo-thermoml sync` on your own machine; the runtime never reads it and the public site has no mirror, so there it shows the install instructions instead. That absence is a status, not an empty result: *"nobody measured this"* and *"you have no mirror"* are different sentences. It is a *reading list, never a value*: no number is copied into a case or a record, and the only outbound action is opening the article. Read it and cite it — never this index.

## 3.15 Evidence (measured points against every candidate)

Pick exactly two components that some measured study is about — methanol and water is the worked example — and the **evidence** lens appears (it is not offered where no dataset for the pair exists). It is the Explorer's answer to *"which model should I trust for this pair?"*, and it answers by showing, never by deciding.

On the left: the **measured datasets** for the pair, gathered from three places — the bundled tutorials' `constant/experiments/`, your private `data/local/evidence/` folder (the development GUI only; the public site never sees it) and the open case — each with its citation, DOI, number of points, pressure and review status. One measurement is one dataset however many places carry a copy: copies are merged by the checksum of their data block, and the list says where else it lives. Below them, the **candidate models**: every parameter record for the pair in any tier (each a separate candidate, labelled by where it came from), UNIFAC where both components carry groups, COSMO-SAC only

where a sigma profile is actually installed, and ideal mixing. Tick and untick both lists; on a phone they fold behind one button.

On the right: the measured bubble temperatures (with the dataset's own uncertainty bars) under one curve per candidate; one **residual** panel per candidate (engine minus measurement); and a table of the mean absolute deviation, its relative form and the bias. A dataset is told by its marker shape, a candidate by its colour. Where a dossier cites a candidate — a pair Choupo regressed in a curation tutorial — its verdict and held-out AAD sit beside it.

*What it computes: nothing physical.* Each candidate is one `choupoProps` run, its parameters written inline, asked *at the measured liquid compositions* (`vary { points ( ...); }`), never read off a curve between two grid points. The view subtracts and averages those answers and fits nothing; fitting and promoting a pair stay with the curation tools. The address carries the selection (`&lens=evidence&datasets=...&models=...`), so the picture a professor pastes is the picture a student gets. Two lessons are waiting in the methanol–water example: the fitted pair is inside the band declared before its fit and still not closer than the predictive model on the held-out study, and part of every residual is the vapour-pressure record's, not the activity model's. **F1** (or the book icon) on this view opens the Theory Guide's section on what is fittable and what is not (`ch:fittable`) — where a parameter comes from, which is what a residual here is evidence about; the view's own *How to read this view* link opens this section.

## 3.16 Phase split along $T$ or $P$ ($V$ / $L$ / $S$, pH)

Pick two or more components the engine can flash — every one carrying a vapour pressure, or all of them components of the open case's own thermophysical system — and the **phase split** lens appears. It answers *"what exists at this temperature, and how much?"* for ONE feed: you give the overall composition in moles, hold $P$ (or $T$) and move $T$ (or $P$) over a range, and at every grid point the engine solves the whole phase split through the flash unit's own solver — vapour, one or two liquids, every crystallising solid a record declares, minerals precipitating where the package speciates. The overall composition stays fixed; what moves is how it is shared between the phases. Cooling a water–ethanol liquor from 0 to $-20\,^{\circ}\mathrm{C}$ shows the freeze-concentration lesson: one liquid, then ice appearing between $-8$ and $-9\,^{\circ}\mathrm{C}$ (a 90/10 molar feed under the curated NRTL pair), then the liquor growing richer in ethanol as more water leaves as ice.

On the left: the **thermophysical system**. *Built here* is a molecular package over the set — the liquid's activity model (ideal, NRTL, Wilson, UNIFAC; under NRTL and Wilson the panel says which pairs a curated record covers) and, for each component whose record declares an enthalpy of fusion and a fusion temperature, a box to let its pure solid crystallise. With a crystal ticked the engine solves liquid + solid with no vapour (a solid beside a vapour is not solved yet), so the phase set is not offered. *The open case's* sends the open case's `constant/` tree as written, its chemistry included — the route by which pH and precipitates reach the lens, since the package built here does not speciate. Below: the **feed** in moles (sent as typed; the engine normalises it to one mole and says so in its own words), the swept variable with its held value, the range and the number of points, and the flash unit's own phase-set word (`VL`, `LL`, `VLLE`, `auto`); for the last three the package built here declares two liquids sharing the one activity model, since the flash refuses to split a liquid it was given only one of.

On the right: the amount of every phase in mol per mol of feed as **stacked bars at the grid points** — the engine's amounts placed on top of each other, summing to one at every solved point; nothing is drawn between two grid points. A phase that **appears or disappears** is marked by a shaded band spanning the two grid values the engine says the change lies between: the boundary is somewhere inside that band, and the view does not claim to know where. Below: **pH** in its own panel where the engine published one (otherwise a sentence saying it did not), and the **composition** of a chosen phase, with a gap where that phase is absent. Then the engine's own sentences: the events, every **refused point** with its reason (a gap in every figure, never a neighbour's value), the record-level announcements at the first point that raised each (a vapour pressure extrapolated below its fitted window, a pair used outside its span), and what the scan does not model.

*What it computes: nothing physical.* One `choupoProps` run of one `phaseSplitScan` operation; every number on screen is a number of its `phaseSplitScans` result block. *What it cannot show:* a solid no record declares. An excipient solidifies here only where its record carries the crystal's data — `sucrose.dat` has no enthalpy of fusion, so sucrose never crystallises in this view — and nucleation and supercooling are not modelled (every point is the equilibrium state). **F1** (or the book icon) opens the Theory Guide's isothermal flash (`ch:flash`), the calculation solved at every point; the view's own *How to read this view* link opens this section.

# 4 Provenance and honesty rules

The Explorer inherits the project's credo:

- **Every number has a source.** Component cards show where each value came from; estimates carry their UNVALIDATED badge; nothing is silently filled in.
- **Disabled means explained.** A view that cannot run tells you which requirement failed (component count, missing VLE data, missing UNIFAC groups) — never a dead button.
- **The Explorer never writes.** It reads the catalogue and the open case; promoting data, fitting parameters and locking goldens belong to the props bench and the curation tools. Sketch here, commit there.

# 5 From Explorer to case: the hand-off

A typical session: you notice in the $T$–$x$–$y$ view that your solvent pair azeotropes at 78 % — so the simple column in your flowsheet cannot deliver 99 % purity. The hand-off is: (1) reproduce the envelope as a props case (`propertyScanBinary`) so the finding is golden-locked; (2) switch the flowsheet to the azeotrope-aware design (pressure-swing or entrainer); (3) cite the Explorer finding in the case header. The Explorer is where you LEARN; cases are where knowledge is kept.